



Multimodal Visual Life Detection using a Compact Tri-Modal Camera Unit

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Abstract

In Search and Rescue (SAR) missions, the ability to rapidly and reliably identify whether a person is alive is critical to saving lives. This thesis investigates the feasibility of detecting life across multiple visual modalities using a compact and cost-efficient tri-modal sensor system—the Compact Tri-Modal Camera Unit for RGBDT Vision (CTCAT)—which captures synchronized RGB, thermal, and depth imagery. The evaluation considers two complementary indicators of life: large-scale body movements, such as walking, crawling, or waving, and subtle chest motion associated with respiratory activity.

To explore this, two real-time detection pipelines were implemented and deployed. The first pipeline uses modality-specific YOLO (You Only Look Once) object detectors combined with multi-person tracking to identify overt human motion. The second extracts respiratory signals by applying dense optical flow to a manually defined thoracic region, allowing the visualization of subtle breathing motion over time. A qualitative evaluation was performed across a range of scenarios varying in environmental conditions, lighting, distance, body posture, and clothing type.

The evaluated scenarios show that RGB imagery is most effective for detecting large-scale movement in well-lit conditions. Thermal imaging complements this capability and serves as the primary modality in low-light environments, both indoors and outdoors. Due to high noise levels and limited range, depth data proved to be the least reliable modality for detecting victim movement and was primarily usable only in close-range indoor scenarios.

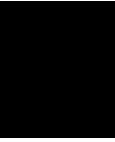
Conversely, subtle chest motion was detectable across all three modalities, with signal quality significantly influenced by camera viewpoint and the surface texture of the victim’s clothing. RGB yielded the clearest breathing signals under good lighting; thermal imaging produced slightly weaker but consistent signals, while depth data was again frequently affected by sensor noise.

These findings demonstrate the feasibility of visual life detection based on both movement and respiration across RGB, thermal, and depth modalities, and highlight the strengths and limitations of each under varying SAR-relevant conditions.

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Introduction

1.1 Motivation

In Search and Rescue (SAR) operations, the timely and reliable detection of victims and their life status is essential to supporting first responders in locating and rescuing victims. In recent years, robotic platforms—such as drones and ground-based robots—have gained increasing attention for their ability to access hazardous or hard-to-reach areas, aiding in victim detection and situational awareness [DMG⁺19, LGJ24].

Visual life detection using computer vision offers a non-contact, real-time approach for identifying individuals and their life status. This thesis explores a specific aspect of this broader challenge: visual life detection using a multimodal sensor system that combines RGB, thermal, and depth imaging. Multimodal approaches are of particular interest because they can leverage complementary characteristics of different sensing modalities—such as visual texture in RGB imagery, heat distribution in low-light scenarios using thermal data, and spatial structure provided by the depth modality—to improve the robustness of detection methods across a range of scenarios.

The focus of this work is on evaluating the feasibility of multimodal life detection using the Compact Tri-Modal Camera Unit for RGBDT Vision (CTCAT)[SK22]. By capturing synchronized data from all three modalities, the sensor enables the investigation of life detection based on both large-scale body movements such as walking, crawling, or waving, and subtle chest motion related to respiration.

Through controlled experiments, this thesis aims to assess the capabilities and limitations of each modality in detecting life signs under different conditions. The results are intended to inform the potential role of an affordable multimodal vision system in supporting life detection tasks relevant to first responders and SAR applications.

1.2 Research Questions

This thesis is guided by two key research aims, which frame the scope and direction of the study.

Objective 1: Life Detection via Large-Scale Movement

Large-scale movement refers to noticeable changes in the position and posture of the body, such as walking, crawling, or waving. These movements are characterized by significant shifts in the body's alignment. This type of movement can serve as an indicator of life, making it a key focus for detection. The aim of this objective is to explore how effectively large-scale victim movement can be detected across different sensing modalities as a proxy for life detection.

- **Movement inference:** Determine whether detection of large-scale movement, such as walking, crawling, or waving, serves as a reliable indicator of life.
- **Modality comparison:** Compare performance of RGB, depth, and thermal modalities in detecting both stationary and moving subjects.
- **Advantages and limitations:** Identify the strengths and weaknesses of this approach under varying operational conditions, including changes in environment, lighting conditions, distance, and number of victims.

Objective 2: Life Detection via Subtle Chest Motion

Subtle chest motion refers to fine-scale movements of the chest wall that occur as a result of respiration and the rhythmic expansion and contraction of the chest region. While these motions are small, they can provide a crucial hint if a person is still breathing and thus alive. This objective aims to assess the feasibility of detecting life signs through these subtle chest motions, without relying on overt victim movement.

- **Detectability of vital signs:** Evaluate whether CTCAT's RGB, depth, and thermal sensors can capture chest-wall displacements corresponding to breathing cycles and pulse.
- **Modality comparison:** Compare the effectiveness of each sensor modality in isolating and measuring subtle chest motions.
- **Advantages and limitations:** Identify the strengths and weaknesses of this approach by testing various combinations of viewing angle, victim position, and subject clothing.

1.3 Structure of the Thesis

The structure of this thesis is outlined below, with each chapter briefly summarized to highlight its primary focus and contribution to the overall investigation.

- **Chapter 2 – Background and Fundamentals:** This chapter outlines the theoretical foundations and contextual background of the thesis. It introduces the principles of RGB, thermal, and depth imaging, presents the Compact Tri-Modal Camera Unit for RGBDT Vision, and provides an overview of the techniques relevant to the two life detection objectives.
- **Chapter 3 – Related Work:** A review of prior research in the fields of search and rescue, victim detection, and visual life detection is provided to contextualize the presented work.
- **Chapter 4 – Methodology:** This chapter describes the two implemented life detection methods—gross body movement detection and subtle chest motion analysis—each applied separately to RGB, thermal, and depth modalities captured by the CTCAT.
- **Chapter 5 – Results and Discussion:** The results are presented and analyzed qualitatively based on a set of predefined evaluation scenarios. The performance and limitations of each modality and life detection method are examined across these scenarios.
- **Chapter 6 – Conclusion:** The final chapter summarizes the key findings, discusses methodological and practical limitations, and outlines potential directions for future research in multimodal visual life detection.

Background and Fundamentals

2.1 Imaging Modalities

RGB imaging captures information from the visible light spectrum across three color channels: red, green, and blue. It remains the most widely used modality in computer vision due to its high spatial resolution and rich texture representation. Moreover, the majority of deep learning models are designed and trained specifically on RGB data for tasks such as image classification, object detection, and semantic segmentation. However, RGB cameras are inherently sensitive to ambient lighting conditions and may exhibit degraded performance in environments with low illumination, high dynamic range, or visual occlusions. This dependence on consistent lighting limits their reliability in scenarios where stable visual input cannot be ensured. [Sze22]

Depth imaging provides pixel-wise distance measurements between the sensor and the observed scene, resulting in three-dimensional structural representations. Depth data is typically obtained through technologies such as structured light, time-of-flight, or stereo vision. Unlike RGB, depth imaging is generally robust to variations in lighting and color, making it particularly effective for capturing shape, motion, and spatial relationships. However, depth sensors often have limitations such as range constraints, susceptibility to reflective surfaces, and noisy imagery. [PLB12]

Thermal imaging operates in the long-wave infrared (LWIR) spectrum, typically ranging from 8 to 14 μm . Unlike visible light, which relies on reflected illumination, LWIR radiation is emitted directly by objects as a function of their surface temperature. Thermal cameras detect this emitted infrared radiation and convert it into temperature-mapped images, enabling passive imaging in low-light conditions or complete darkness. This makes thermal data particularly valuable for applications such as heat detection, human presence monitoring, and non-contact physiological measurement. Nonetheless, thermal cameras typically offer lower spatial resolution compared to RGB or depth sensors. [VM18]

2.2 Compact Tri-Modal Camera Unit

The Compact Tri-Modal Camera Unit for RGBDT Vision (CTCAT) [SK22] is a compact, cost-efficient sensor system (approximately 450 USD, as of November 2021) designed to capture synchronized RGB, depth, and thermal images. At its core, the CTCAT builds upon the Orbbec Astra¹ RGBD camera, which combines a standard RGB camera and a structured light depth sensor. These two modalities are augmented by a compact FLIR Lepton 3.5² thermal camera, integrated into a custom 3D printed housing to complete the tri-modal setup. An overview of each sensor’s specifications is provided in Table 2.1

Table 2.1: Technical specifications of the CTCAT imaging components (adapted from Table 1 in [SK22])

	RGB	Depth	Thermal
Device	Orbec Astra	Orbec Astra	FLIR Lepton 3.5
Technology	–	Structured Light	Microbolometer
Resolution	1280×720	640×480	160×120
Frame rate	30 fps	30 fps	8.7 fps
Operating range	0 to ∞	0.6–8m	0 to ∞
Wavelength	Visible	$0.83\mu\text{m}$ (NIR)	8–14 μm (LWIR)
FoV (H, V, D)	($60^\circ, 49.5^\circ, 73^\circ$)	($60^\circ, 49.5^\circ, 73^\circ$)	($57^\circ, 42^\circ, 71^\circ$)
Accuracy	–	$\pm 1\text{cm}@3\text{m}$	$\pm 5K$
Data Format	8-bit	16-bit (depth in mm)	14-bit (temperature in K)

To ensure temporal and spatial consistency, all RGB, depth, and thermal frames are undistorted, geometrically aligned, and rescaled to a uniform resolution of 640×480 pixels. Aligned frame triplets can be used directly or saved lossless as 16-bit PNGs for further processing. The system’s maximum synchronized frame rate is limited to 8.7 fps due to the thermal camera, which also determines the capture rate for RGB and depth to maintain synchronization.

The CTCAT is equipped with an NVIDIA Jetson Orin³, a single-board computer used to run the real-time life detection implementations developed in this thesis.

In addition, the CTCAT serves as the primary data acquisition device for capturing multimodal recordings used in the evaluation scenarios presented in later chapters.

¹<https://web.archive.org/web/20211014122034/https://orbbec3d.com/product-astra-pro/>, accessed July 12, 2025.

²<https://www.flir.com/products/lepton/>, accessed July 12, 2025.

³<https://www.nvidia.com/de-de/autonomous-machines/embedded-systems/jetson-orin/>, accessed July 12, 2025.

2.3 Computer Vision Methods

2.3.1 YOLO

You Only Look Once (YOLO) [RDGF16] is a real-time object detection algorithm that performs classification and localization in a single forward pass through a Convolutional Neural Network (CNN). Unlike two-stage detectors such as Faster R-CNN [RHGS17], which first generate region proposals and then classify them, YOLO adopts a one-stage architecture that directly predicts bounding boxes and associated class probabilities from the input image. This unified structure enables high-speed inference while maintaining competitive accuracy, making it particularly suitable for time-sensitive applications. Since its original introduction in 2016, the YOLO architecture has undergone multiple iterations, improving in speed, accuracy, and model efficiency. This thesis uses the YOLO11n model, a lightweight and fast variant, implemented via the Ultralytics⁴ Python package.

2.3.2 ByteTrack

ByteTrack [ZSJ⁺22] is a real-time multi-object tracking algorithm designed to improve the robustness of detection-based tracking systems. It extends traditional tracking-by-detection frameworks by leveraging both high- and low-confidence detections, which are typically discarded in conventional approaches. This inclusive strategy allows ByteTrack to maintain a more consistent identity tracking over time, particularly in challenging conditions involving occlusions, motion blur, or missed detections. The algorithm operates by applying motion prediction (via a Kalman filter) and solving a frame-to-frame assignment problem using the Hungarian algorithm. In this thesis, ByteTrack is used in combination with YOLO-based object detection to associate victim detections across video frames, enabling temporal analysis of large-scale body movement in RGB, depth, and thermal video streams.

Together, YOLO and ByteTrack are used to detect and track victims in RGB, depth, and thermal videos. Significant changes in tracked positions over time are interpreted as indications of movement and, consequently, potential signs of life.

2.3.3 Farneback Dense Optical Flow

Optical flow characterizes the apparent motion of image points, or pixel intensities, between consecutive frames in a video sequence, representing how objects or regions move over time [Sze22]. The Farneback algorithm, introduced in 2003 [Far03], estimates dense optical flow by modeling each local neighborhood with a polynomial function and computing a displacement field that covers the entire image. This dense representation captures fine-grained motion patterns and is particularly effective for detecting small, smooth movements. Due to its balance between computational efficiency and accuracy,

⁴<https://github.com/ultralytics/ultralytics>, accessed July 12, 2025.

Farneback flow is well-suited for applications involving subtle physiological motion. In this thesis, the standard OpenCV implementation⁵ of the algorithm is employed to analyze motion in the chest region. Subtle vertical displacements over time are interpreted as breathing activity, providing a non-contact method for respiratory life detection across RGB, depth, and thermal modalities.

⁵https://docs.opencv.org/3.4/d4/dee/tutorial_optical_flow.html, accessed July 12, 2025.

Related Work

The integration of robotic systems into SAR missions has emerged as a key technological advancement, offering novel capabilities that complement and extend the reach of human first responders. Recent survey studies have highlighted the evolution of rescue robotics from teleoperated ground vehicles to increasingly autonomous and multi-agent systems capable of operating in complex, dynamic environments. Rescue robots operate across a variety of environments, each presenting unique challenges and opportunities for deployment. Ground-based robots, for example, are commonly used in search scenarios, where they navigate through rubble and confined spaces to inspect structures and detect victims at close range. In contrast, marine and amphibious systems play a critical role in flood response and underwater missions, assisting in coastal or submerged environments where terrestrial systems are ineffective. Aerial drones offer high mobility and a broad field of view, making them particularly useful for surveying large or inaccessible areas quickly, such as disaster zones with obstructed ground access. [LGJ24, DMG⁺19, QTCP⁺20].

3.1 Victim Detection

Victim detection is a core task in SAR operations, and numerous approaches have been proposed to address this challenge. Focusing on computer vision systems, recent research has explored a range of methods for visually detecting victims in disaster-stricken environments [HPS19, CUPSBDC21, CUOdCB24, SK18]. Deep learning-based approaches, particularly those using CNNs, have become a dominant trend due to their ability to generalize across complex visual conditions. A notable example for this is Hartawan et al. [HPS19] who propose a CNN-based victim detection system implemented on a lightweight computing platform. Their approach utilizes a MobileNet-SSD model to classify and localize human victims in camera streams. The study demonstrates the feasibility of deploying compact, high-performance CNN models for real-time victim

detection in field scenarios, offering a cost-effective and portable solution for SAR applications.

In another study, Cruz Ulloa et al. [CUPSBDC21] apply deep learning models—including YOLO [RDGF16], Fast R-CNN [Gir15], and SSD [LAE⁺16]—to thermal imagery for victim detection in post-disaster scenarios with low visibility. Their system, mounted on a mobile ground robot, is trained and evaluated in visually cluttered scenes and achieves high detection accuracy using thermal data alone. Expanding on this work, Cruz Ulloa et al. [CUOdCB24] present a comparative evaluation of RGB, thermal, and multispectral imaging modalities under controlled indoor and outdoor SAR conditions. The results demonstrate that RGB, thermal, and multispectral imagery each exhibit distinct strengths depending on the operational context, underscoring the value of multimodal sensing for robust performance in complex and challenging environments.

Whereas the majority of existing approaches focus on RGB and thermal imaging for victim detection, depth information remains underutilized in SAR research. A notable exception is the work by Saputra and Kormushev [SK18], who propose a geometric method for detecting human bodies using 3D point cloud data from an on-board RGB-D sensor. Their approach projects the point cloud onto the ground plane and analyzes the resulting grid map to identify human-like shapes.

However, these studies all focus solely on victim detection rather than inferring life status. In the course of researching for this thesis, no study was found that systematically compares RGB, thermal, and depth modalities for victim detection in SAR environments while also leveraging movement-based cues as indicators of life.

3.2 Non-Contact Life Sign Detection

While gross body movement is a useful indicator of human presence, many victims in SAR scenarios may be unconscious, motionless, or trapped in ways that prevent visible movement. In such cases, non-contact life sign detection—especially respiration monitoring—offers a critical extension to victim detection methods. Instead of focusing on static posture or overt motion, these techniques aim to infer subtle physiological activity, such as chest motion from breathing, through visual analysis.

Recent developments have shown that contactless respiratory monitoring is feasible using standard or low-cost imaging systems. Massaroni et al. [MSS⁺18] propose a method that estimates respiratory rate and breathing pattern by analyzing light intensity fluctuations in video recordings of the upper chest region using a standard RGB camera. A follow-up comparison study [MSSM19] evaluates two different approaches—pixel intensity tracking and optical flow—and concludes that the optical flow method achieved superior accuracy.

Janssen et al. [JWMdH15] introduce an automated, video-based respiration monitoring system that detects the respiratory region of interest without manual input. Their method relies on motion factorization to isolate respiratory-induced chest or abdominal motion as an independent component from the video signal. Tested on a dataset of 148

video sequences, the method achieves high agreement with reference standards such as guided breathing and ECG.

Extending the application of contactless respiration monitoring to mobile SAR platforms, Orbea et al. [OUCB24] developed a legged-manipulator robot capable of detecting respiratory motion in post-disaster environments. The system combines CNN-based victim identification with optical flow-based motion analysis, using an RGB camera mounted on the manipulator’s end effector. Once a victim is localized, the robot approaches the thoracic region and estimates the respiratory rate based on detected chest motion. Notably, the system is tested in both indoor and outdoor environments and shows robustness to difficult lighting conditions and diverse clothing textures, which are common obstacles in field scenarios. This work emphasizes the potential for mobile, non-invasive robotic platforms to support rapid triage and medical evaluation in SAR operations.

Together, these approaches illustrate the growing feasibility of vision-based respiration monitoring, with promising results in both clinical and unstructured environments. However, current systems typically rely on a single sensing modality and often lack direct comparisons across different environmental conditions, clothing types, and postures.

Methodology

This study investigates the real-time life detection capabilities of the CTCAT across multiple modalities, employing two complementary approaches: (1) detecting life signs through large-scale body movements, and (2) detecting life signs at close range via subtle chest motion.

Each objective is addressed through a dedicated detection pipeline designed to operate in real time on a single-board computer—specifically, the NVIDIA Jetson Orin, which is attached to the CTCAT.

The full source code of this thesis is available on GitHub, as referenced in Appendix B.

4.1 Life Detection via Large-Scale Movement

The large-scale movement-based life detection pipeline identifies deliberate overt human motions—such as walking, crawling, gesturing, or limb movement—as indicators of presence and vitality.

Three separate YOLO [RDGF16] object detectors—each fine-tuned on modality-specific subsets of the SDT [PSK20] and TRISTAR [SHK24] datasets—detect human subjects in every incoming frame. To enable temporal association and distinction between victims, detected individuals are subsequently tracked using ByteTrack [ZSJ⁺22], which links detections across consecutive frames to generate continuous bounding-box trajectories. To determine movement, the system analyzes temporal variations in each bounding box’s centroid position and area. If the displacement or shape change exceeds predefined thresholds, the subject is classified as “moving,” indicating potential signs of life.

Figure 4.1 provides an overview of the complete detection pipeline for this objective, illustrating the process from initial frame acquisition through victim detection and tracking to the final movement-based decision logic.

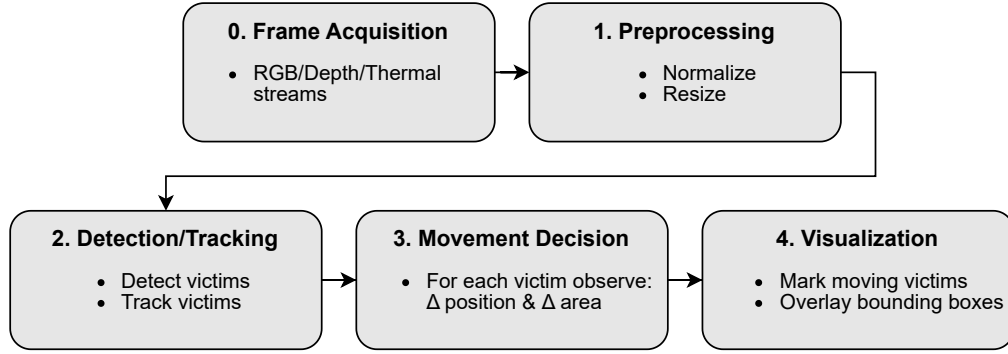


Figure 4.1: Pipeline: Life Detection via Large-Scale Movement.

4.1.1 Model Training

Given the requirement for real-time performance on the CTCAT-integrated single-board computer, the lightweight YOLO11n variant from the Ultralytics library is selected. Ultralytics provides a YOLO11n model pretrained on the COCO [LMB⁺14] dataset, which serves directly for our RGB modality. However, no publicly available YOLO11n checkpoints exist for thermal or depth imagery, which are essential for our use case. To address this limitation, modality-specific YOLO11n models are initialized from the pretrained RGB detector and subsequently fine-tuned using transfer learning.

Custom Dataset Composition

The combined dataset draws on the publicly available TRISTAR[SHK24] and SDT[PSK20] datasets. TRISTAR comprises 15,618 registered RGB, depth, and thermal frames, each captured from multiple viewpoints and environments. Because it targets human segmentation and action detection, TRISTAR provides only segmentation masks and temporal action labels rather than YOLO-formatted bounding boxes; to tackle this, a custom Python script is used, which converts each segmentation mask into the corresponding YOLO bounding box annotation.

SDT contributes 40,000 synthetically rendered depth and thermal images and 8,000 real-world captures of humans in various postures—lying, sitting, and standing—each supplied with ground-truth bounding boxes. Especially the high number of lying individuals in this dataset is crucial for our SAR-specific environments, where victims are often found lying on the ground.

From both datasets, the designated training split is used for model training and the validation split for performance evaluation, resulting in a total of 41,959 training images and 10,848 validation images for our custom dataset. The idea of combining the real-world captures of TRISTAR with the synthetically generated imagery of SDT is to increase the number of training images and also positional diversity, thereby improving model robustness to the wide range of victim configurations encountered in SAR scenarios.

Table 4.1 summarizes the details of the TRISTAR, SDT, and the combined custom dataset. Unfortunately, both datasets remain limited to indoor scenes, highlighting the need for future inclusion of outdoor or mixed-environment imagery to fully leverage the potential of these models in diverse real-world settings. However, collecting and annotating such training images lies beyond the scope of this thesis and is therefore left for future work.

Table 4.1: Comparison of the datasets by split and image modality.

Dataset	Split	Thermal Images	Depth Images
TRISTAR	Training	9959	9959
	Validation	2848	2848
	Test	2811	2811
SDT	Training	32000	32000
	Validation	8000	8000
	Test	8000	8000
Combined	Training	41959	41959
	Validation/Test	10848	10848

Fine-Tuning

Each modality-specific YOLO11n detector for depth and thermal is fine-tuned using the combined SDT [PSK20] and TRISTAR [SHK24] corpus. The task is formulated as a single-class detection problem with the class label `person`, as the system focuses exclusively on detecting victims and their corresponding bounding boxes to enable downstream movement analysis. Fine-tuning is conducted on an NVIDIA A30 GPU with 24 GB of VRAM using the following hyperparameters: a batch size of 64, an initial learning rate of 0.01 with a cosine learning rate scheduler, weight decay of 0.0005, and a total of 50 training epochs. To enhance model robustness and generalization, all default data augmentation techniques provided by the Ultralytics library are applied during training.

Figure 4.2 and Figure 4.3 show the training curves of the depth- and thermal-specific YOLO11n models over 50 epochs. The plots display training and validation losses for bounding box regression, object classification, and Distribution Focal Loss (DFL), along with key performance metrics including precision, recall, and mean Average Precision (mAP). DFL is a loss function designed to improve bounding box localization by modeling the target as a probability distribution over discretized coordinates, rather than as a single scalar value. The mAP, or mean Average Precision, summarizes detection performance by averaging precision across classes and Intersection-over-Union (IoU) thresholds. Specifically, the plots include mAP@0.5 and mAP@0.5:0.95 as standard evaluation metrics. For both models, training and validation losses steadily decrease and converge by epoch 50, indicating stable learning with no signs of overfitting. Likewise,

mAP@0.5 approaches 1.0, while mAP@0.5:0.95 exceeds 0.9 by the final epoch, with no notable divergence between training and validation curves.

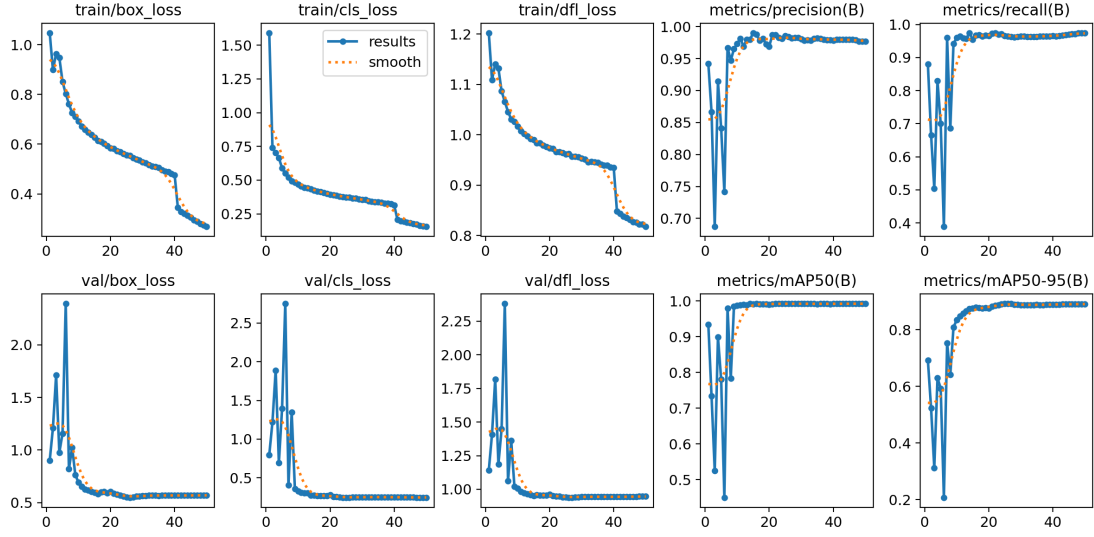


Figure 4.2: Depth model training: loss and mAP curves over 50 epochs.

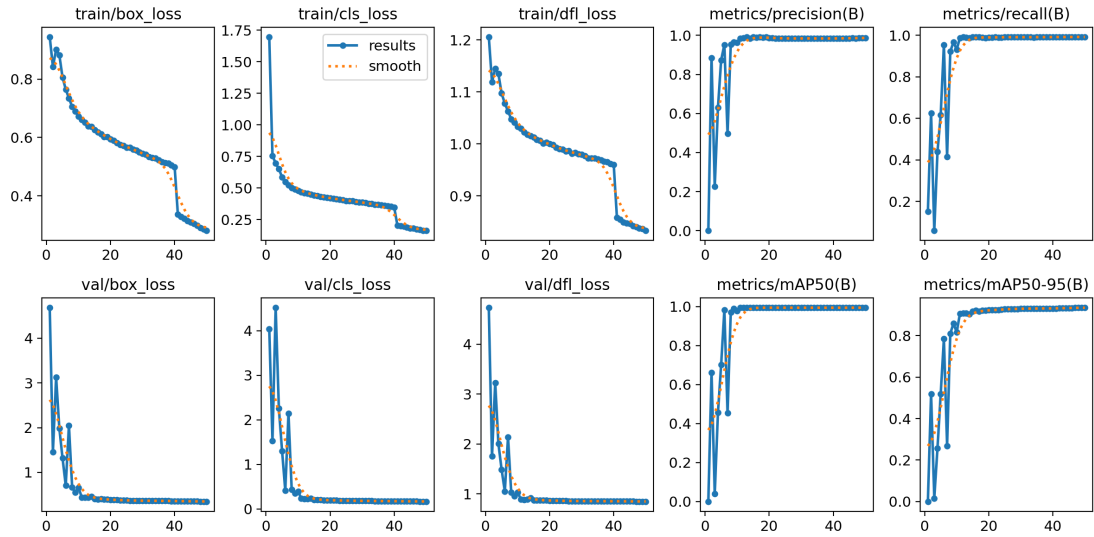


Figure 4.3: Thermal model training: loss and mAP curves over 50 epochs.

4.1.2 Detection Pipeline

The previously fine-tuned models are now employed in our movement detection pipeline to identify and classify overt human motion in different modalities. The following sequence of operations is applied to every frame in real time:

1. **Preprocessing** The raw depth and thermal frames arrive as 16-bit encoded single-channel image arrays $I_{16}(x, y)$ with intensities in $[0, 65535]$. As YOLO expects 8-bit encoded 3-channel images, each pixel value is then linearly rescaled into an 8-bit $[0-255]$ range by the following formula:

$$I_8(x, y) = \frac{I_{16}(x, y)}{65535} \cdot 255 \quad (4.1)$$

After scaling, the single-channel result I_8 is converted to a three-channel RGB image by replicating the gray value across each channel. RGB frames bypass this step and are used directly without modification. Manual resizing is not needed as this is done implicitly by the YOLO model.

2. **Victim Detection and Multi-Victim Tracking.** The preprocessed frame is passed to the fine-tuned YOLO11n model operating in “track” mode for object detection and subsequent tracking with ByteTrack. Each detected victim is assigned a bounding box and a persistent track ID, enabling consistent identity tracking across frames and reducing fragmentation during partial occlusions. This allows the system to distinguish between multiple victims and assess their individual motion status, helping to identify which victim may require urgent assistance.
3. **Movement Decision.** For each detected victim, the pipeline computes motion based on the change in position and area of the bounding box across consecutive frames. The movement metric is computed relative to the size of the bounding box to account for scale invariance in different perspectives or distances. Let (c_x, c_y) denote the center of the current bounding box and (c'_x, c'_y) the center in the previous frame. Let A and A' be the respective areas. The relative center displacement Δ_{pos} and relative area change Δ_{area} are computed as:

$$\Delta_{\text{pos}} = \frac{\sqrt{(c_x - c'_x)^2 + (c_y - c'_y)^2}}{\sqrt{w^2 + h^2} + \varepsilon} \quad (4.2)$$

$$\Delta_{\text{area}} = \frac{|A - A'|}{A' + \varepsilon} \quad (4.3)$$

where w and h are the width and height of the current bounding box, and ε is a small constant to avoid division by zero.

A victim is considered to be moving in a given frame if either Δ_{pos} or Δ_{area} exceeds their respective thresholds, τ_{pos} and τ_{area} , defined empirically (default: $\tau_{\text{pos}} = 0.03$, $\tau_{\text{area}} = 0.05$).

To reduce the effect of noise and sporadic false positives, an additional temporal debouncing mechanism is employed. A fixed-length history window of recent motion decisions is maintained for each track ID. A victim is classified as “moving” only if at least k out of the last n frames (default: $k = 3$, $n = 5$) exceed the motion threshold.

4. **Visualization.** To visualize detected victims and indicate their motion status, the live video feed is annotated in real time. Depending on the selected modality (RGB, depth, or thermal), frames are first color-mapped for improved perceptual clarity. Detected victims are then highlighted with bounding boxes: green for moving individuals and red for stationary ones. The fully annotated stream is rendered at the sensor’s native frame rate and displayed either locally or remotely, enabling operators to perform immediate situational assessments. Representative sample frames for each modality are shown in Figure 4.4.

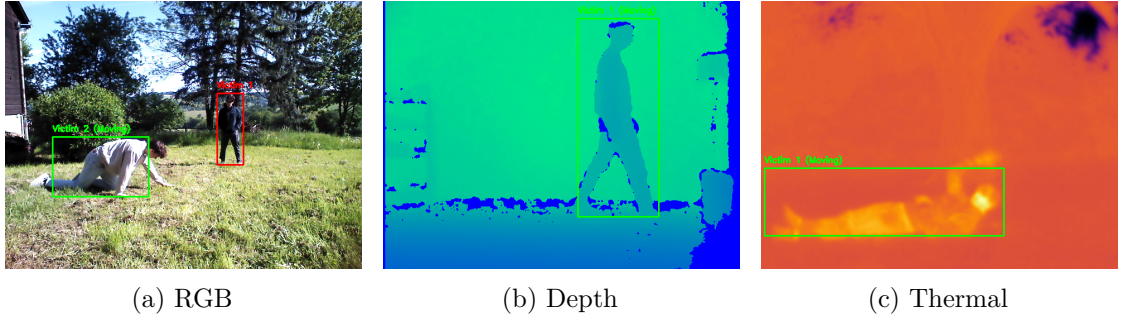


Figure 4.4: Life Detection via Large-Scale Movement. Moving persons are marked in green, whereas motionless victims are denoted red.

4.2 Life Detection via Subtle Chest Motion

In scenarios where overt, large-scale movement is absent—such as when victims are unconscious or immobilized due to injury—our first approach alone may fail to indicate signs of life. To overcome this limitation, a second detection pipeline focuses on subtle thoracic displacements associated with respiration. The approach begins with the operator manually defining a rectangular Region-of-Interest (ROI) over the subject’s chest in the initial frame. For each subsequent frame pair, optical flow is computed to estimate pixel-wise motion. The vertical component of the flow within the ROI is then spatially aggregated to yield a one-dimensional respiratory signal over time. By analyzing the periodicity and amplitude of this signal, a victim’s vitality might be inferred. Figure 4.5 illustrates the outline of this approach.

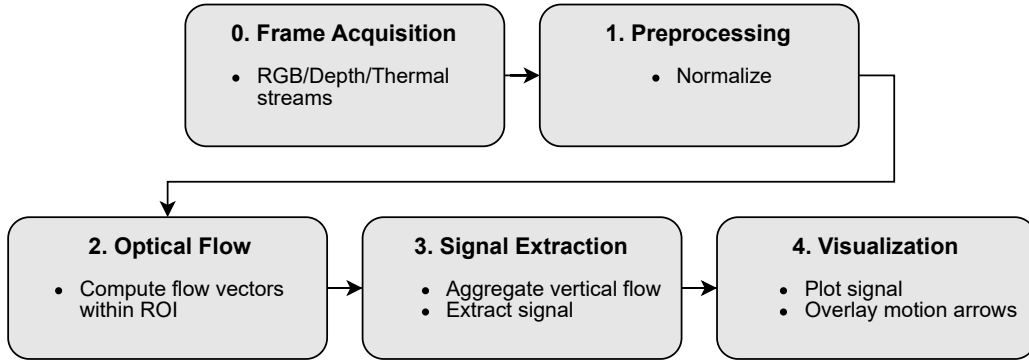


Figure 4.5: Pipeline: Life Detection via Subtle Chest Motion.

4.2.1 Detection Pipeline

After manually selecting a ROI over the victim’s thoracic area during initialization, each frame once again undergoes a series of processing steps:

1. **Preprocessing.** Again, the first step is to convert the raw 16-bit depth or thermal frames $I_{16}(x, y)$ given by the sensor to 8-bit grayscale. This time, however, Min-Max scaling is used instead of the static linear normalization applied in the large-scale movement detection pipeline. This normalization is performed as follows:

$$I_8(x, y) = \frac{I_{16}(x, y) - \min(I_{16})}{\max(I_{16}) - \min(I_{16})} \times 255 \quad (4.4)$$

where **min** and **max** denote the per-frame minimum and maximum pixel intensities. This dynamic scaling enhances the contrast within each frame, thereby improving the accuracy of optical flow estimation by amplifying subtle local variations. RGB frames again bypass this step and are used directly without modification.

2. **Optical Flow.** In this step of the pipeline, optical flow vectors are computed between the previous and the current frame using Farneback’s dense optical flow algorithm [Far03]. OpenCV’s implementation of this algorithm is used, with the following empirically determined parameters:

- `pyr_scale = 0.5`
- `levels = 3`
- `winsize = 15`
- `iterations = 3`
- `poly_n = 5`
- `poly_sigma = 1.2`
- `flags = 0`

This results in a dense two-dimensional flow field $\mathbf{F}^{(t)}$, which encodes the horizontal and vertical motion vectors for each pixel at time step t . The vertical component within the previously defined ROI is then used in the subsequent steps to extract a respiratory signal.

3. **Signal Extraction.** Let $\mathbf{F}_{\text{ROI}}^{(t)}$ denote the subarray of the optical flow field $\mathbf{F}^{(t)}$ corresponding to the previously defined ROI. From this, we extract the vertical flow component $v^{(t)}(x, y)$ at each pixel $(x, y) \in \text{ROI}$ and compute the average vertical motion as

$$s^{(t)} = \frac{1}{|\text{ROI}|} \sum_{(x,y) \in \text{ROI}} v^{(t)}(x, y) \quad (4.5)$$

where $|\text{ROI}| = w \times h$ is the total number of pixels in the ROI. The resulting scalar value $s^{(t)}$ is appended to a cumulative signal vector $\{s^{(1)}, s^{(2)}, \dots\}$, forming a time series of aggregated vertical flow values.

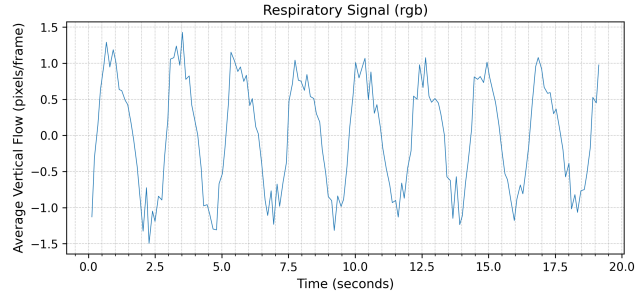
Since respiration induces quasi-periodic vertical expansion and contraction of the thorax, the signal $s^{(t)}$ typically oscillates around zero. By analyzing the temporal evolution of this signal—particularly its periodicity and amplitude—it is possible to infer the presence of respiratory motion and thus determine whether the subject is alive.

4. **Visualization.** To visualize respiratory motion and highlight subtle chest movement, the live video feed is annotated in real time. Depending on the selected modality (RGB, depth, or thermal), each frame is again color-mapped for optimal visibility. A rectangle delineates the ROI over the thoracic area. Semi-transparent arrows representing the Farneback flow vectors are overlaid on a regular grid within the ROI so that vertical displacements due to breathing become apparent. Simultaneously, a separate window displays the average vertical-flow signal, updating

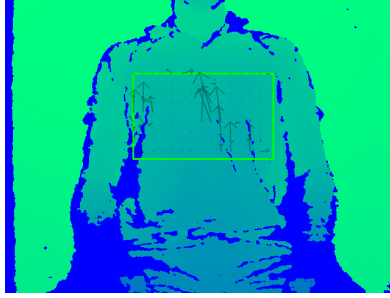
in real time so that the respiratory activity of the victim can be observed. Both the annotated video stream and the evolving respiratory waveform render at the sensor's native frame rate, allowing operators to immediately assess whether the subject's chest is rising and falling. Representative frames and corresponding signal plots for each modality are shown in Figure 4.6.



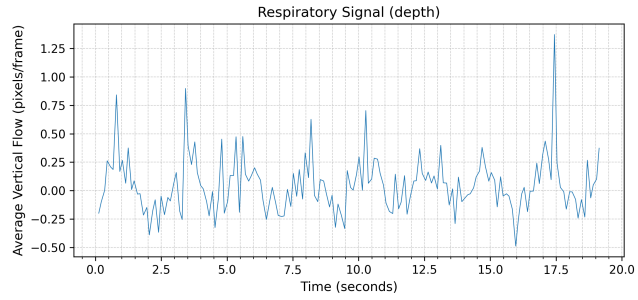
RGB – Flow Overlay



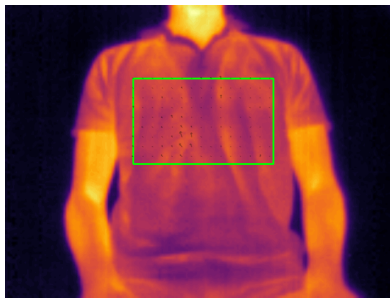
RGB – Respiratory Signal



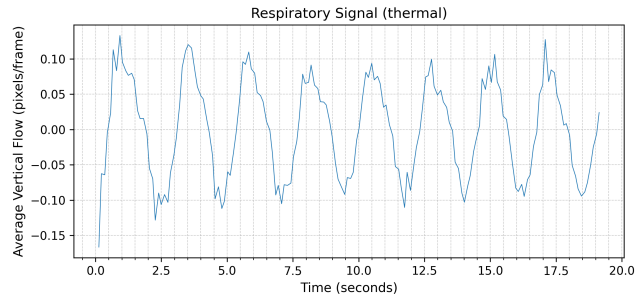
Depth – Flow Overlay



Depth – Respiratory Signal



Thermal – Flow Overlay



Thermal – Respiratory Signal

Figure 4.6: Life Detection via Subtle Chest Motion. Left: Optical flow vectors overlaid on RGB, depth, and thermal images. Right: Corresponding respiratory signal extracted over time.

Results and Discussion

The evaluation of the two proposed multimodal life detection approaches is conducted qualitatively due to the diversity and complexity of real-world SAR scenarios, as well as the absence of standardized or directly comparable datasets. Selected test scenarios are recorded and analyzed under varying environmental conditions to assess the strengths and limitations of each method in each modality and to identify specific situations where the approaches perform reliably or encounter challenges.

5.1 Life Detection via Large-Scale Movement

Eight distinct scenarios are designed (see Table 5.1), systematically varying key conditions such as environment, distance, lighting, and the number of individuals to detect. Across these scenarios, participants perform a range of movement types and adopt various body positions. Representative frames from the recordings are shown in Figure 5.1. Supplementary sample evaluation sequences for each modality are also provided in Appendix A.1 to illustrate the evaluation process further.

5.1.1 Modality Performance Across Scenarios

Large-scale movements such as walking, crawling, and waving are generally reliable indicators of life when victims are clearly detected. However, the quality and robustness of detection vary significantly across the three sensing modalities and under different environmental conditions. In the following analysis, the RGB, depth, and thermal modalities are examined separately across all scenarios to highlight the specific strengths and limitations of each.

Scenario	Distance	People	Environment	Lighting
S1	3.5m	1	Indoor	Day
S2	3.5m	1	Indoor	Night
S3	3.5m	1	Outdoor	Day
S4	3.5m	1	Outdoor	Night
S5	7m	2	Indoor	Day
S6	7m	2	Indoor	Night
S7	7m	2	Outdoor	Day
S8	7m	2	Outdoor	Night

Table 5.1: Life Detection via Large-Scale Movement: Scenario Overview.



Figure 5.1: Example frames from near distance (top row) and far distance (bottom row) scenarios captured during daylight and nighttime (colorized)

RGB

RGB imaging consistently yields the most accurate and stable results across all daylight scenarios (S1, S3, S5, S7). Detection is reliable in both indoor and outdoor settings, effectively identifying and tracking human subjects across various body positions and movement types. Movement classification is particularly robust: the system consistently distinguishes between static and moving individuals, with almost no false positives and no missed movements, assuming the victim’s bounding box is detected. Even subtler movements like waving are consistently detected and accurately classified as motion. Also, multiple-person scenarios are handled well, maintaining stable detection and movement tracking across different configurations. The availability of rich visual features under daylight conditions, combined with the robustness of the object detection model—likely due to its pretraining on a wide variety of images as described in the methodology section—enables precise spatial localization, making RGB particularly effective for recognizing posture and movement.

However, RGB performance declines under conditions with reduced visual contrast—such

as when victims are lying in grassy environments, positioned further from the camera, or not upright in outdoor settings—where detection occasionally fails due to background similarity (see Figure 5.2). Additionally, the modality is inherently limited by its dependence on ambient lighting: in low-light or nighttime scenarios, the RGB sensor is unable to detect any subjects, making it ineffective without external illumination. These limitations highlight the need for integrating complementary sensing modalities in search and rescue applications.

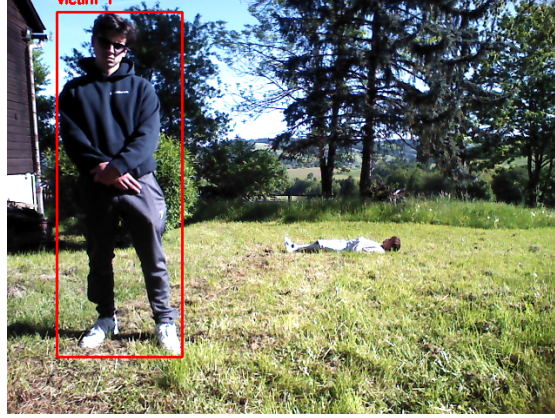


Figure 5.2: Scenario S7: Missed detection due to textural similarity.

Depth

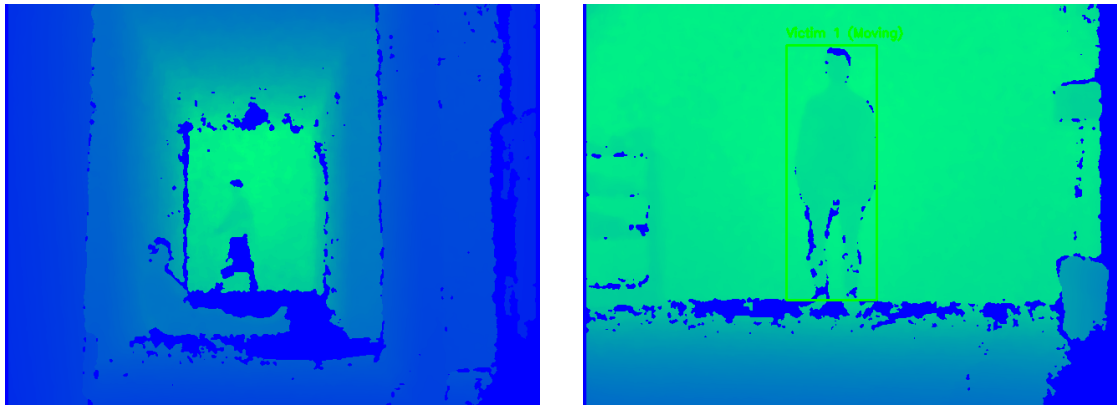
The depth modality shows highly context-dependent performance, proving useful in selected scenarios but facing significant limitations in many others. Its best results can be observed in indoor, near-range settings (e.g., Scenario S2), where it reliably detects gross movements such as walking and crawling. However, in the other tested scenarios, movement detection only works partially, and performance quickly deteriorates under less controlled conditions.

The most severe limitations occur in outdoor environments, where the CTCAT’s depth sensor is unusable in daylight. Direct sunlight causes strong infrared interference, completely saturating the sensor and preventing reliable operation. Even at night, when the sensor could technically function outdoors, detection remains unreliable. Complex natural environments—such as grass, bushes, or uneven terrain—obscure the subject or introduce background noise, leading to missed detections or false positives.

Indoors, the sensor also struggles in certain configurations. Detection frequently fails when subjects lie on the ground or lean against walls, as the depth sensor has difficulty distinguishing body contours from flat or textured surfaces. In narrow hallways or cluttered areas, the depth signal becomes unstable and noisy, impairing both detection and tracking (see Figure 5.3a). This instability also affects movement classification: bounding boxes around stationary individuals exhibit jitter or small fluctuations due to

inconsistent depth readings, which leads to misclassification of still subjects as moving (see Figure 5.3b).

A fundamental limitation of the depth modality is its restricted operational range of 8 meters (see Table 2.1), which significantly constrains its applicability in real-world SAR scenarios. At increased distances, human silhouettes become less well-defined and more difficult to distinguish from surrounding objects, thereby impairing the performance of the employed object detection algorithm. Even within the nominal 8-meter range, detection accuracy deteriorates noticeably, confining the sensor’s utility to close-range applications.



(a) Scenario S6: No detection due to noise and similarity to wall and background.

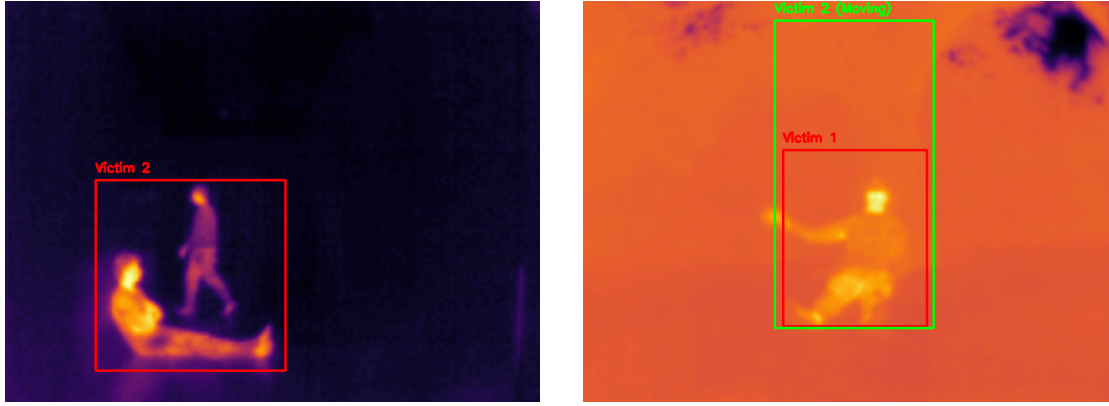
(b) Scenario S2: Victim marked as "moving" because of unstable detection.

Figure 5.3: Typical failure cases in depth sensing, including instability, missed detections, and misclassifications

Thermal

With the multitude of limitations with depth imagery, thermal imaging becomes the primary modality under low-light or night-time conditions. The large-scale movement detection approach performs particularly well in indoor environments, as the thermal heat signature—and thus also the contours—of victims and other objects are easily distinguishable for the employed object detector. Even at greater distances, victims are correctly detected and classified as either still or moving, as the thermal sensor of the CTCAT provided sufficient resolution and contrast in these conditions.

However, thermal sensing also exhibits several modality-specific limitations, many of which appear to stem from the characteristics of the trained detection model rather than the thermal sensor itself. In cluttered scenes—such as narrow indoor hallways or outdoor environments with closely spaced individuals—the detector struggles to differentiate between adjacent subjects, leading to overlapping detections or merged detections and confusion in motion attribution (shown in Figure 5.4a).



(a) Scenario S6: Merged detection. Overlapping victim detections merge to one bounding box.

(b) Scenario S4: False positive double detection. A tree is incorrectly identified as a victim due to its similar thermal signature.

Figure 5.4: Typical failure cases in thermal sensing, including merged detections and false positives caused by background heat patterns

Additionally, in outdoor conditions, the effectiveness of thermal sensing is reduced by environmental heating. Sun-exposed surfaces like grass and trees decrease the temperature contrast between victims and the background, leading to missed detections or incorrect movement classification. While thermal generally performs better than the depth modality when victims lie on the floor or lean against objects, it still shows instability in scenes where ambient temperatures create contours similar to human shapes. This is particularly noticeable in scenarios with trees and bushes, which cause false positives or even double detections (see Figure 5.4b). While thermal imaging is not without drawbacks, it remains essential for low-light conditions in search and rescue and offers valuable support when RGB and depth are limited.

5.1.2 General Limitations of the Approach

Despite showing promising results in several cases (see, for example, Appendix A.1), the movement-based life detection approach exhibits fundamental limitations that constrain its applicability in real-world SAR operations.

A core limitation lies in its reliance on observable motion as a proxy for life. If a victim is unconscious, severely injured, or intentionally remaining still, the system lacks any direct means of inferring the presence of life. As such, the method is inherently conservative: it provides strong evidence of life when motion is detected but cannot offer any assurance in the absence of movement. Another key constraint is the assumption of a stationary sensor platform. Since movement detection relies on temporal variations in bounding box position and size, camera motion interferes with this process and invalidates the results, unless more sophisticated motion compensation methods are implemented.

The overall effectiveness of the approach is also tightly coupled to the accuracy and

robustness of the underlying object detector. For the RGB modality, the used state-of-the-art pretrained model, which is trained on a large and diverse dataset, enables reliable victim detection across a wide range of conditions. In contrast, the thermal and depth modalities rely on custom fine-tuned detectors trained on a comparatively limited dataset, consisting only of near-range indoor environments. This reduces their generalization capabilities and leads to a multitude of issues, including double detections, ID switching in temporal victim tracking due to inconsistent, false, lost, or merged detections, and frequent false positive movement classifications. These issues are particularly apparent in the thermal and depth modalities, where the object detectors are notably less stable.

In summary, while large-scale movement detection offers an operationally relevant signal for life inference, its reliability is fundamentally constrained by factors such as posture sensitivity, environmental conditions, sensor placement, detection accuracy, and the inherent ambiguity of using motion as an indirect indicator of life.

5.2 Life Detection via Subtle Chest Motion

12 scenarios are designed to systematically explore the performance of the chest motion analysis across a range of different configurations. As the effects of distance and lighting conditions are already addressed in the previous approach, these scenarios focus on the influence of victim posture, camera angle, and clothing type. The complete scenario configuration is summarized in Table 5.2. Across all scenarios, subjects maintain a static pose while breathing normally, breathing rapidly, or holding their breath—simulating apnea—allowing the system to isolate and analyze the victim’s respiratory motion. Representative example frames from the recorded scenarios are shown in Figure 5.5. Additional sample evaluation sequences for each modality are provided in Appendix A.2.

Scenario	Body Position	Clothing	Angle
S1	Sitting	Shirt	frontal
S2	Sitting	Sweater	frontal
S3	Sitting	Jacket	frontal
S4	Sitting	Shirt	side-view
S5	Sitting	Sweater	side-view
S6	Sitting	Jacket	side-view
S7	Leaning	Shirt	frontal
S8	Leaning	Sweater	frontal
S9	Leaning	Jacket	frontal
S10	Lying	Shirt	top-down
S11	Lying	Sweater	top-down
S12	Lying	Jacket	top-down

Table 5.2: Life Detection via Subtle Chest Motion: Scenario Overview.

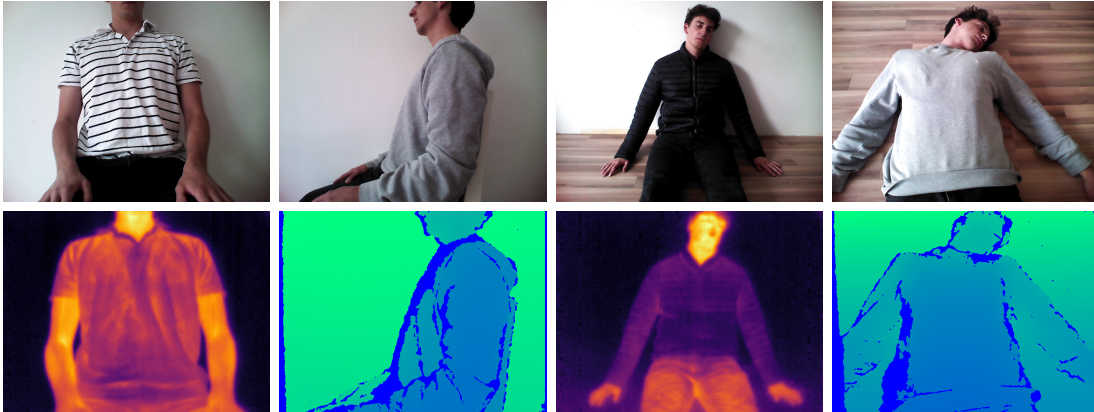


Figure 5.5: Example frames from chest motion detection scenarios. The grid shows variations across body positions, clothing types, and camera distance/angle.

5.2.1 Respiratory Signal Characteristics

Figure 5.6 presents a representative respiratory signal plot for one of the recorded scenarios. Each recording typically spans approximately 80 seconds and is captured and analyzed across all three modalities. The plot illustrates three distinct breathing phases: simulated apnea from 0–20 seconds, normal breathing from 20–50 seconds, and accelerated breathing from 50–80 seconds.

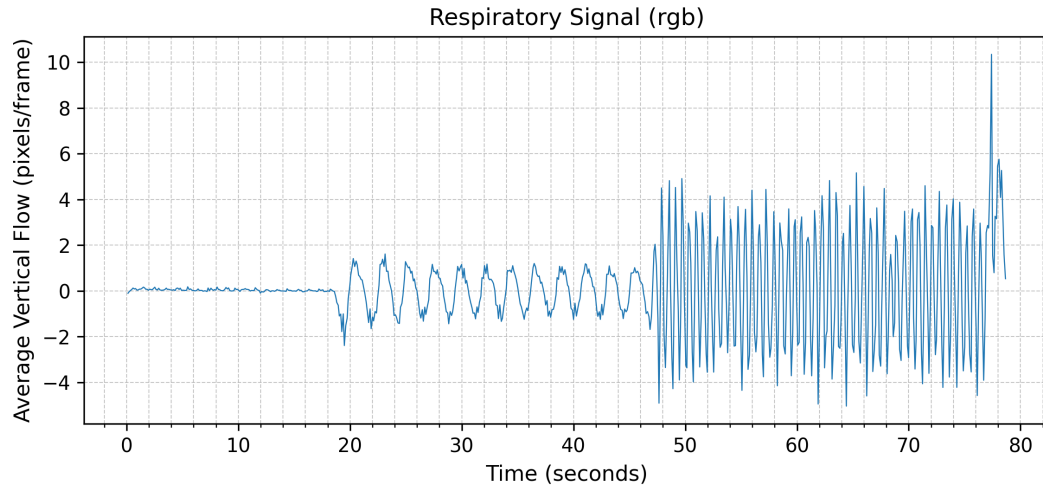


Figure 5.6: Signal plot of a subject in a seated position (frontal view, wearing a shirt).

The signal appears as an oscillatory curve, with both amplitude and frequency reflecting the underlying breathing activity. From this signal, three key indicators can be extracted: (1) the presence or absence of periodic oscillations, indicating the existence of breathing activity; (2) the frequency of these oscillations, which provides an approximate respiratory rate (breaths per minute)—higher frequencies correspond to faster breathing; and (3) the amplitude of the oscillations, which reflects the strength of the registered chest motion and enables differentiation between shallow and deep breathing.

5.2.2 Modality Performance Across Scenarios

The three modalities are evaluated independently across all scenario configurations. In some scenarios, respiratory activity is detectable across all modalities, while in others, certain modalities struggle to reliably extract a meaningful signal.

RGB

The RGB modality demonstrates the most consistent and reliable performance across all 12 evaluated scenarios. Optical flow-based motion analysis in the RGB channel produces

high-quality respiratory signals characterized by strong periodicity, low noise, and clear phase separation between apnea, normal, and accelerated breathing.



Figure 5.7: RGB modality showing clear vertical flow arrows.

This robustness stems from the high spatial resolution and rich visual texture available in the RGB images. Clothing with structured patterns (e.g., shirts or jackets) provides strong visual cues that enhance motion vector clarity. Even in cases where clothing texture is less distinct—such as with plain sweaters—shadows, folds, and lighting variations introduce sufficient surface features for the algorithm to track chest movement effectively. As mentioned earlier, all scenarios used for the chest motion analysis were captured under daylight conditions.

Performance is best in frontal views (Scenarios S1–S3), where a large surface area of the chest is visible. Even in side views (S4–S6), despite a smaller visible region, motion signals remain clearly detectable, though with slightly reduced amplitude. Leaning poses (S7–S9) and top-down lying positions (S10–S12) also yield stable signals, albeit with modest amplitude attenuation due to reduced optical flow magnitude caused by viewing angle effects.

Figure 5.7 shows an overlay of motion vectors corresponding to the signal plot in Figure 5.6. In all RGB scenarios, Farneback dense optical flow produces clear and coherent vectors that predominantly point in vertical directions, accurately reflecting the upward and downward movement of the chest during respiration.

Depth

The depth modality again exhibits the most unstable and inconsistent performance across the evaluation. While certain configurations enable the detection of periodic breathing signals, the modality struggles significantly under most conditions due to sensor noise, limited surface detail, and depth quantization errors.

In optimal frontal views (see Appendix A.2), the sensor can capture subtle chest surface movements, particularly when subjects wear form-fitting clothing with pronounced 3D texture, which introduces measurable depth variations. However, these signals are still contaminated by high-frequency noise and spurious spikes in the motion vectors.

Side views (S4–S6) and leaning postures (S7–S9) substantially reduce the size and visibility of the chest region, making the respiratory signal nearly indistinguishable from random fluctuations. The reduced size introduces even more noise in the depth imagery. Also in top-down lying poses (S10–S12), the system has difficulties extracting a clear signal as the viewing angle leads to poor depth resolution and a flat, noisy signal.

Clothing also plays a critical role. While shirts with natural wrinkles sometimes provide enough 3D structure to generate meaningful signals, sweaters and jackets with smoother or looser fits often lead to flat or erratic signals lacking periodicity. In many of these cases, it becomes difficult to distinguish between actual breathing and random noise, especially during apnea phases, where the signal appears equally unstructured. Figure 5.8 presents a typical scenario where the depth modality struggles. The signal exhibits erratic amplitude and lacks a clear periodic structure. The motion vectors are small due to the low 3D structure, but with relatively large outliers, overshadowing the breathing motion signal.

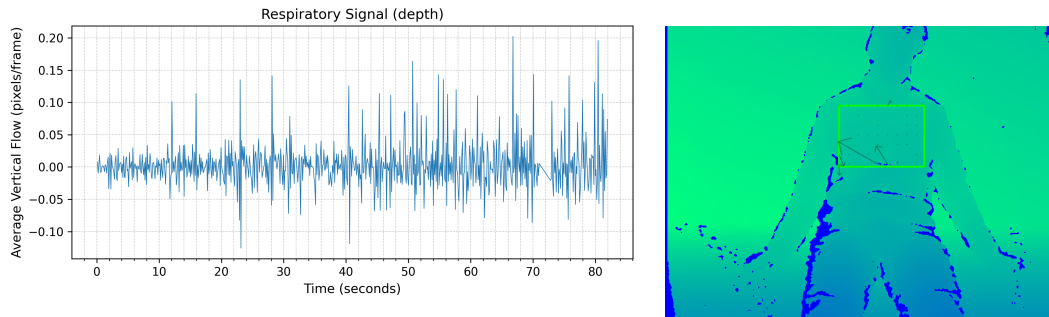


Figure 5.8: Depth modality: Victim in a leaning position, frontal view, wearing a jacket.

Thermal

The thermal modality provides consistently interpretable respiratory signals across all evaluated conditions, with performance lying between that of RGB and depth. While the absolute signal amplitude is generally lower than that observed in RGB, the thermal sensor offers stable performance across clothing types and postures. Although not explicitly tested in these scenarios, the thermal modality would likely serve as the primary sensing method in situations where lighting conditions are insufficient for reliable RGB imaging.

Unlike RGB and depth, thermal sensing is largely unaffected by visual appearance or surface texture, instead relying on temperature distribution and subtle thermal changes caused by chest expansion and contraction. As such, it remains reliable even when the

subject wears flat, textureless, or loose-fitting clothing such as sweaters or jackets. This makes it especially valuable in scenarios where RGB motion tracking struggles due to a lack of visual cues.

In frontal sitting positions (S1–S3), the signal is clear and periodic, allowing for robust differentiation between apnea, normal, and accelerated breathing. Side views and leaning postures (S4–S9) result in slightly reduced contrast due to the smaller visible area and less pronounced temperature variation. In top-down lying poses (S10–S12), signal quality remains interpretable, although the amplitude difference between normal breathing and accelerated breathing states is sometimes less distinct (see Figure 5.9).

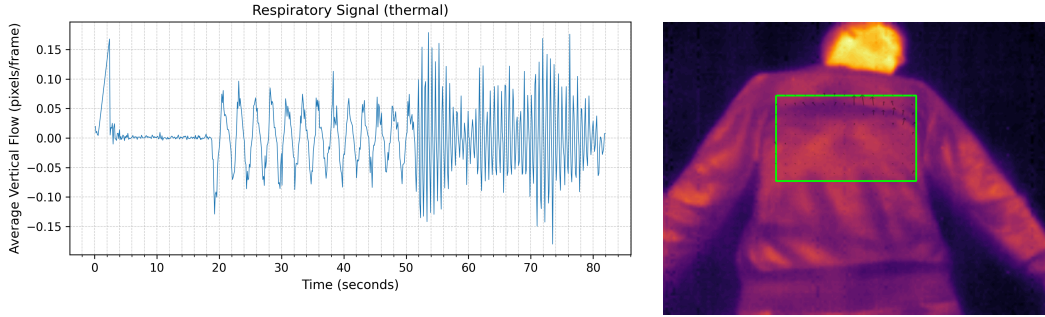


Figure 5.9: Thermal modality: Victim in lying position, frontal view, wearing sweater

5.2.3 General Limitation of the Approach

While the chest motion-based life detection technique demonstrates strong potential in controlled settings, its effectiveness is limited by several inherent challenges when applied to real-world scenarios.

First and foremost, the method is inherently constrained to near-range use. The respiratory signal is extracted from pixel-level optical flow within a defined Region-of-Interest (ROI) on the chest. As the distance between the camera and the subject increases, the apparent size of the chest in the image diminishes, reducing the number of pixels available for motion analysis. This leads to a weaker and less detailed signal, as the spatial resolution within the ROI is no longer sufficient to reliably capture subtle breathing-induced motion.

The approach also requires a stationary camera setup. Because respiratory motion is detected based on small, localized temporal variations in pixel displacement, any movement of the camera introduces global motion artifacts that can easily overwhelm or distort the signal.

A further constraint lies in the modality-agnostic nature of the optical flow computation. The current method applies a standard two-dimensional optical flow algorithm across all input modalities, treating each as a flat grayscale image regardless of the underlying data characteristics. This is especially limiting for depth data, which inherently encodes

three-dimensional spatial information. By ignoring the depth modality’s true 3D nature and reducing it to a 2D image, the system fails to leverage its unique advantages, resulting in unstable and noisy signals compared to RGB or thermal inputs.

In addition, the relatively low frame rate of the CTCAT (see specifications Table 2.1) imposes a significant technical limitation. With a synchronized acquisition rate of approximately 8.7 frames per second across all three modalities, the temporal resolution of the recorded breathing motion is coarse. This sparse sampling restricts the system’s ability to accurately capture the frequency and subtle dynamics of respiratory activity, especially during faster or irregular breathing patterns. A higher frame rate would allow for finer signal granularity and improve the temporal precision of breathing phase transitions.

Overall, while the method proves effective in highlighting subtle chest motion in controlled near-range settings, its broader applicability remains constrained by camera placement, sensor resolution, frame rate, and the lack of modality-specific motion modeling.

Conclusion

This thesis demonstrated that life can be reliably detected in real time using computationally efficient techniques by analyzing both large-scale body movements as well as subtle respiratory chest motion, although detection effectiveness varied depending on the sensing modality and environmental conditions.

6.1 Key Findings

For detecting large-scale movements, RGB imaging proved to be the most reliable, effectively identifying movement in most recorded scenarios. It performed well in both indoor and outdoor environments, although it sometimes struggled in visually cluttered settings, where the background and the victims appeared similar. Thermal imaging emerged as the primary modality for low-light environments, where RGB was ineffective. When detections were stable, movement recognition was also reliable, and individual tracking was possible. The main challenges for thermal imaging arose when objects with similar heat distribution and shape to humans were present, leading to false positives. Depth modality proved to be the most unreliable, grappling with the intrinsic noisiness of the depth sensor. This made detection difficult when victims were lying on the floor or near walls, where noise and background could cause the victim to blend in or be misidentified. Additionally, depth struggled in outdoor scenarios, particularly in direct sunlight, where sensor interference reduced its effectiveness.

In detecting subtle chest motion caused by respiration, RGB and thermal modalities again outperformed depth. RGB imaging captured subtle chest displacements with clarity, especially in frontal and side views, where the chest was more visible. Thermal imaging successfully detected temperature variations associated with chest movement, making it particularly useful in low-light conditions. In contrast, depth sensing struggled with subtle chest motion, especially at greater distances or when chest movement was

minimal. Its noisy signals and limited resolution hindered its ability to accurately detect these fine motions.

In conclusion, RGB and thermal modalities were most effective for both large-scale movement detection and subtle chest motion detection, with each modality excelling under different conditions. Depth imaging, while sometimes effective in controlled indoor environments, proved less suitable for SAR tasks due to its limited range, noise, and environmental sensitivity. Combining these modalities offers the most robust solution for life detection in SAR scenarios, with the potential to overcome the individual limitations of each sensor type.

6.2 Limitations

Despite the promising outcomes, several limitations of the study must be acknowledged. First, the analysis relies on a qualitative approach, as ground truth data for neither of the two tested life detection approaches was available in all three modalities. While the results provide valuable insights into the system’s behavior, the findings are not quantitatively validated. For the large-scale movement detection approach, an evaluation against manually labeled ground truth annotations of movement presence would enable the use of classification metrics such as accuracy, precision, recall, and detection latency (time until movement is detected). In contrast, for the subtle chest motion approach, the extracted motion signal could be quantitatively assessed by comparing the estimated respiratory rate to that obtained via external reference methods, such as contact-based respiration sensors or manual breath counts.

The methodological approaches selected for large-scale movement detection and subtle chest motion analysis each come with inherent limitations. The bounding box method used for large-scale movement detection performs reliably in well-structured environments but is fundamentally dependent on the precision of the underlying object detection model. Any misdetections or tracking failures can directly compromise movement inference. Conversely, the optical flow-based approach for subtle chest motion detection relies on accurate estimation of pixel-wise displacement. This method is sensitive to image resolution and sensor noise, particularly when the subject is positioned at a greater distance. In such cases, fine motion details may be lost, reducing the method’s effectiveness.

The sensor limitations of the CTCAT also play a significant role in the performance of these approaches. The low frame rate (approximately 8.7 fps) limits the temporal resolution of motion detection, making it difficult to capture fast or irregular movements, particularly for subtle chest motion. Additionally, the depth sensor faces challenges with range and resolution, especially in outdoor environments where sunlight or complex geometry introduces significant interference.

6.3 Future Work

While this thesis demonstrated the potential of multimodal life detection using the CTCAT, several avenues for improvement remain. A key direction is to explore sensor fusion by combining RGB, depth, and thermal modalities. Developing ways to integrate all modalities in one fusion model could enhance detection accuracy and robustness, compensating for the individual weaknesses of each modality in complex environments.

Regarding detection methods, the current bounding box approach for large-scale movement detection could be improved with alternative techniques, such as skeletal tracking or more advanced motion analysis. While the optical flow-based approach proved effective for RGB data, it could be replaced or complemented by modality-specific methods that better exploit the characteristics of thermal and depth imaging, such as leveraging absolute temperature values in thermal data or distance measurements in the depth modality.

Additionally, the study relied on static camera setups, but dynamic SAR environments often require moving imaging platforms. Future work should focus on adapting a system for moving camera setups to handle camera movement in real-world applications, such as drone-based or handheld SAR operations.

In summary, future research should aim to advance multimodal sensor fusion, enhance detection methods, and adapt the system to dynamic environments to improve its practical applicability in SAR operations.

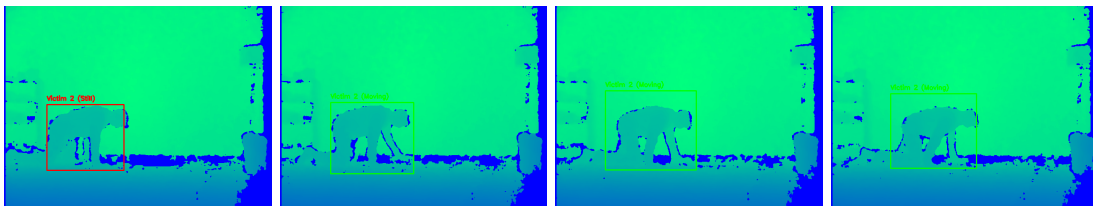
Sample Scenario Sequences

A.1 Life Detection via Large-Scale Movement

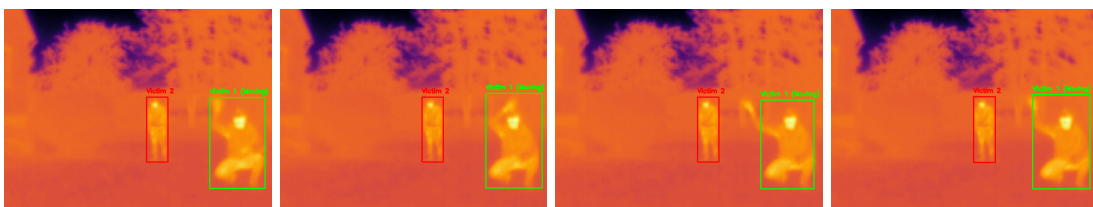
Additional representative sequences where the large-scale movement-based life detection approach performed particularly well across RGB, depth, and thermal modalities.



RGB Modality



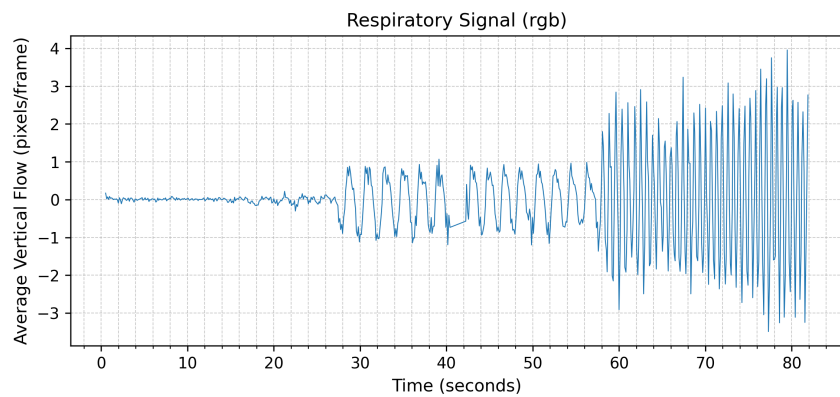
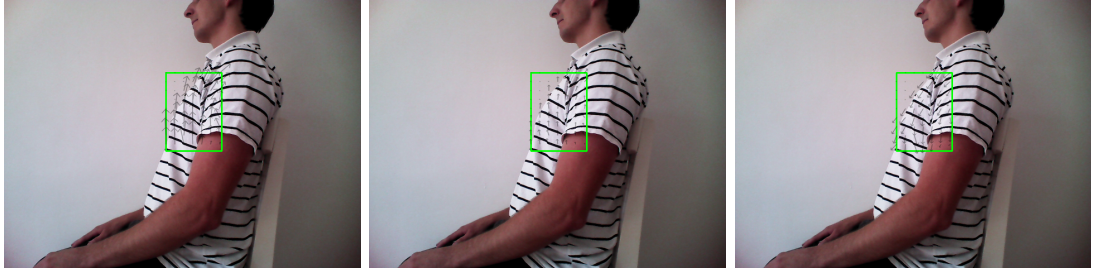
Depth Modality



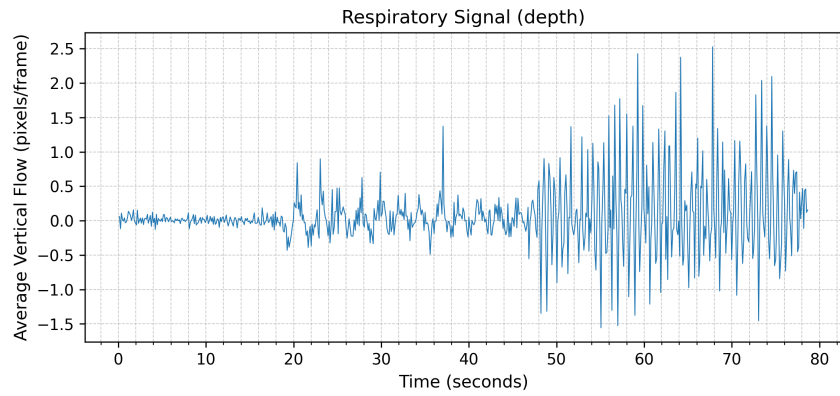
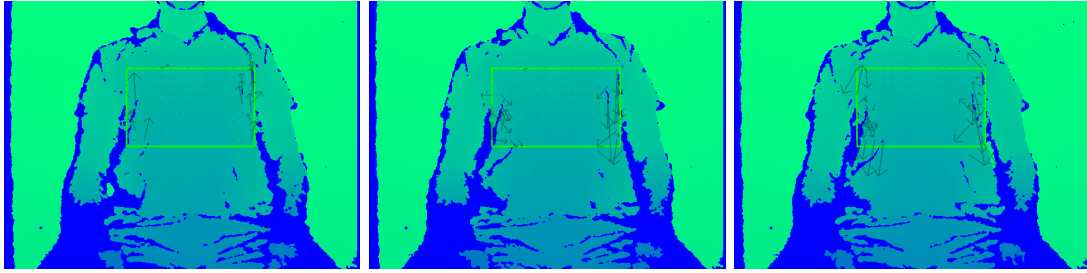
Thermal Modality

A.2 Life Detection via Subtle Chest Motion

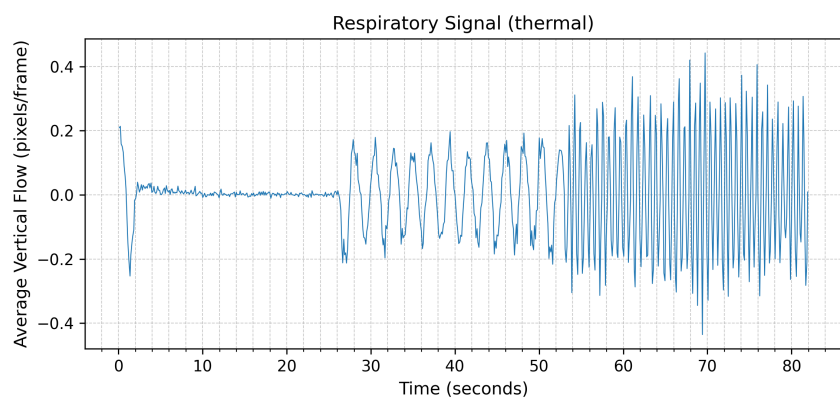
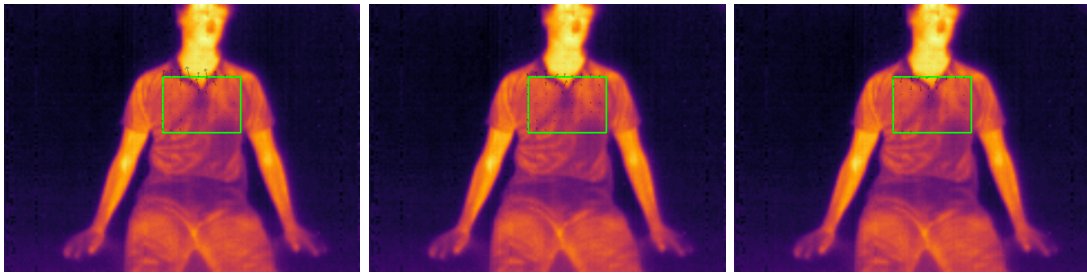
Further example sequences illustrating effective performance of the chest motion-based life detection approach across RGB, depth, and thermal modalities.



RGB Modality



Depth Modality



Thermal Modality

Code and Data

This thesis is supported by a publicly accessible codebase to ensure transparency and facilitate reproducibility of all presented results.

B.1 Code Repository

The complete source code — including both real-time life detection systems, the trained YOLO models, and all evaluation scripts — is available at:

- **GitHub Repository:** <https://github.com/niklasgruener/Life-Detection>

Overview of Generative AI Tools Used

ChatGPT was used to rephrase and improve the clarity, coherence, and structure of selected sections of the text during the writing process. In addition, Grammarly was employed to identify and correct spelling, punctuation, and grammatical errors. No AI tools were used to generate original content or ideas.

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Acronyms

CNN Convolutional Neural Network. 7, 9, 11

CTCAT Compact Tri-Modal Camera Unit for RGBDT Vision. 1–3, 6, 13, 14, 25, 26, 34, 36, 37, 49

DFL Distribution Focal Loss. 15

IoU Intersection-over-Union. 15

LWIR long-wave infrared. 5

mAP mean Average Precision. 15, 16

ROI Region-of-Interest. 19, 20, 33

SAR Search and Rescue. 1, 9–11, 14, 23, 26, 27, 36, 37

YOLO You Only Look Once. 7, 10, 13, 14, 17, 43

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